

Dr. SKA Notes

IV Fluids

Physiology, Fluid Types, Treatment Protocols, Indications, Contraindications, Advantages & Disadvantages

A clinical quick-reference summary. This document is for educational purposes and should be used alongside institutional protocols, current guidelines, and clinical judgment. It is not a substitute for prescribing information or individualized patient assessment.

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1. Body Fluid Physiology Overview

Total body water (TBW) is approximately 60% of body weight in adult men and ~50% in adult women (higher in infants, lower in the elderly and in obesity). TBW is distributed between two major compartments:

Compartment	% of TBW	% of Body Weight	Key Ions / Notes
Intracellular fluid (ICF)	~66%	~40%	High K ⁺ , Mg ²⁺ , phosphate; low Na ⁺
Extracellular fluid (ECF)	~33%	~20%	Subdivided into plasma and interstitial fluid
— Plasma (intravascular)	~8% of ECF	~5%	High Na ⁺ , Cl ⁻ , HCO ₃ ⁻ ; contains proteins (albumin) that exert oncotic pressure
— Interstitial fluid	~25% of ECF	~15%	Similar electrolyte composition to plasma, minimal protein

Only IV fluids given directly expand the intravascular space initially; where fluid ultimately distributes depends on its tonicity and the permeability of vessel walls to water, electrolytes, and large molecules (proteins/starches).

2. Fluid Compartments & Movement

Osmolality vs. Tonicity

Osmolality is the total solute concentration of a fluid (mOsm/kg), including both permeant (e.g., urea, glucose) and impermeant solutes. **Tonicity** refers only to the effect a fluid has on cell volume — determined by impermeant solutes (mainly Na⁺ and its accompanying anions) that cannot freely cross cell membranes.

Isotonic fluids (~275–295 mOsm/L) stay largely in the extracellular space and do not shift water into or out of cells.

Hypotonic fluids cause water to shift into cells (risk of cerebral edema). Hypertonic fluids pull water out of cells into the extracellular space.

Starling Forces (Capillary Fluid Exchange)

Fluid movement between plasma and interstitium is governed by the balance of hydrostatic pressure (pushes fluid out of capillaries) and oncotic pressure from plasma proteins, chiefly albumin (pulls fluid into capillaries). Hypoalbuminemia (e.g., liver disease, nephrotic syndrome, malnutrition, critical illness) lowers oncotic pressure and promotes third-spacing/edema, which is a key rationale sometimes cited for colloid use, though outcome evidence is mixed (see Section 5).

Distribution After IV Infusion (Approximate)

Fluid Type	Fraction Remaining Intravascular at ~1 hr	Typical Clinical Effect
Isotonic crystalloid (e.g., 0.9% NaCl, Lactated Ringer's)	~20-25%	Most distributes to interstitium; ~3-4x the infused volume needed to achieve the same intravascular expansion as colloid
D5W (free water once glucose metabolized)	~8-10%	Distributes across all body water compartments including ICF

Fluid Type	Fraction Remaining Intravascular at ~1 hr	Typical Clinical Effect
Colloid (albumin, starches)	~70-100% initially	Stays intravascular longer due to oncotic effect (duration varies by agent)

3. Classification of IV Fluids

IV fluids are broadly divided into **crystalloids** (water with small dissolved solutes such as electrolytes or glucose that pass freely through capillary membranes) and **colloids** (contain larger molecules — proteins or synthetic polymers — that do not readily cross capillary membranes and exert oncotic pressure).

- **Crystalloids:** Normal saline (0.9% NaCl), Lactated Ringer's / Hartmann's, Plasma-Lyte / Normosol (balanced solutions), D5W, D5NS, D5 1/2NS, hypotonic saline (0.45% NaCl), hypertonic saline (3%, 5%, 7.5%, 23.4%)
- **Colloids:** Human albumin (5%, 25%), hydroxyethyl starches (HES — largely restricted now), gelatins, dextrans, fresh frozen plasma (blood product, not a pure volume expander)

4. Crystalloids — Composition, Uses, Pros/Cons

0.9% Normal Saline (NS)

Composition	Na+ 154, Cl- 154 mEq/L; Osm ~308 mOsm/L (isotonic, but non-physiologic Cl- load)
Indications	Hypovolemia, hemorrhagic shock (initial), hyponatremia, burns, diluent for blood products/medications, metabolic alkalosis, DKA initial resuscitation
Contraindications / Caution	Large-volume resuscitation risk of hyperchloremic metabolic acidosis; caution in renal impairment, heart failure, edematous states
Advantages	Compatible with blood products; widely available; useful when chloride-responsive alkalosis or hyponatremia present
Disadvantages	Supraphysiologic chloride can cause hyperchloremic acidosis, may worsen renal perfusion/AKI risk with large volumes

Lactated Ringer's (LR) / Hartmann's Solution

Composition	Na+ 130, K+ 4, Ca2+ 2.7 (or 3), Cl- 109, lactate 28 mEq/L; Osm ~273 mOsm/L (mildly hypotonic, 'balanced')
Indications	General resuscitation, surgical/perioperative fluid, burns (Parkland formula), trauma, most general hypovolemia
Contraindications / Caution	Avoid/caution with hyperkalemia (contains K+), severe liver failure (impaired lactate metabolism), avoid mixing directly with blood products in same line (calcium can cause clotting) or with some drugs (e.g., ceftriaxone), caution in hypercalcemia
Advantages	Lower chloride load than NS, less risk of hyperchloremic acidosis, more physiologic electrolyte profile, lactate acts as a bicarbonate precursor
Disadvantages	Contains K+ and Ca2+ so not ideal in renal failure/hyperkalemia; incompatible with certain drugs; lactate may confound serial lactate measurements in sepsis

Plasma-Lyte / Normosol (Balanced Crystalloids)

Composition	Na+ 140, K+ 5, Mg2+ 3, Cl- 98, acetate 27, gluconate 23 mEq/L; Osm ~294 mOsm/L; calcium-free
Indications	Similar to LR; preferred in some ICU/sepsis and renal protocols; safe to co-administer with blood products (calcium-free)
Contraindications / Caution	Caution in severe hepatic impairment (acetate metabolism); more costly than NS/LR
Advantages	Most physiologically balanced electrolyte profile; calcium-free (blood-compatible); may reduce AKI/acidosis risk vs NS in some large trials
Disadvantages	Higher cost; less universally available than NS/LR

D5W (5% Dextrose in Water)

Composition	Isotonic in the bag (~252 mOsm/L) but becomes free water once dextrose is metabolized — physiologically hypotonic
Indications	Free water replacement, hyponatremia, vehicle for medications, hypoglycemia treatment
Contraindications / Caution	Avoid as primary resuscitation fluid in hypovolemic/hemorrhagic shock; avoid/caution in cerebral edema, avoid in hyponatremia; caution in diabetics (hyperglycemia)
Advantages	Provides free water and some calories; useful for correcting hyponatremia
Disadvantages	Poor intravascular volume expander (~90% distributes out of vessels quickly); risk of hyponatremia and cerebral edema if overused; hyperglycemia risk

Hypotonic Saline (0.45% NaCl, 'Half-Normal')

Composition	Na+/Cl- 77 mEq/L; Osm ~154 mOsm/L
Indications	Maintenance fluid (with added K+/dextrose per protocol), hyponatremia, free water deficit replacement
Contraindications / Caution	Avoid in hyponatremia, burns, trauma, or acute resuscitation; caution with increased intracranial pressure (risk of cerebral edema)
Advantages	Provides both electrolytes and free water for maintenance needs
Disadvantages	Poor choice for volume resuscitation; risk of hyponatremia and cerebral edema, especially in children

Hypertonic Saline (3%, 5%, 7.5%, 23.4%)

Composition	3% NaCl: Na+/Cl- 513 mEq/L; Osm ~1030 mOsm/L (highly hypertonic)
Indications	Symptomatic/severe hyponatremia, cerebral edema, elevated ICP (traumatic brain injury), refractory shock (small-volume resuscitation in select protocols)
Contraindications / Caution	Requires central line for concentrated forms (peripheral for short 3% infusions per protocol); avoid rapid correction of chronic hyponatremia (osmotic demyelination syndrome risk); caution in heart failure, renal failure
Advantages	Rapidly restores plasma osmolality/serum sodium; effective at reducing ICP; small volumes needed
Disadvantages	Risk of osmotic demyelination if corrected too fast; hyponatremia, fluid overload, phlebitis with peripheral administration, requires frequent electrolyte monitoring

5. Colloids — Composition, Uses, Pros/Cons

Human Albumin (5%, 25%)

Composition	5% is roughly iso-oncotic; 25% is hyperoncotic and pulls fluid from interstitium into plasma
Indications	Spontaneous bacterial peritonitis (with paracentesis), large-volume paracentesis, hepatorenal syndrome, select burn resuscitation, hypoalbuminemia with refractory edema, some sepsis protocols after crystalloid
Contraindications / Caution	Caution in severe heart failure/fluid overload, severe anemia; avoid in patients with known allergy to albumin products
Advantages	Stronger and longer-lasting intravascular volume expansion per mL than crystalloid; useful oncotic effect in hypoalbuminemia
Disadvantages	Expensive; blood-derived product (small infection/allergy risk); no consistent mortality benefit over crystalloid in general resuscitation (per major trials, e.g., SAFE study)

Hydroxyethyl Starches (HES) & Gelatins

Composition	Synthetic colloids; various molecular weights and substitution ratios
Indications	Historically used for volume expansion in surgery/trauma; now largely avoided in critically ill/septic patients
Contraindications / Caution	Avoid in sepsis, critical illness, renal impairment, or bleeding disorders — associated with increased risk of acute kidney injury and coagulopathy; many regulatory agencies restrict or have withdrawn HES use in these populations
Advantages	Effective intravascular expansion, historically cheaper than albumin
Disadvantages	Increased AKI risk (notably shown in sepsis trials such as 6S and CHEST), coagulopathy, pruritus, largely fallen out of favor in critical care

6. Choosing a Fluid by Clinical Scenario

Scenario	Preferred Fluid(s)	Avoid
General hypovolemia / dehydration	Balanced crystalloid (LR, Plasma-Lyte) or NS	D5W as sole resuscitation fluid
Hemorrhagic shock / trauma	Balanced crystalloid bridge; early blood products (massive transfusion protocol) if major bleeding	Large-volume crystalloid alone in major hemorrhage; hypotonic fluids
Sepsis / septic shock	Balanced crystalloids first-line (per Surviving Sepsis Campaign); albumin may be added in large-volume resuscitation	Starches (HES); excess NS (hyperchloremic acidosis)
DKA	NS initially, then transition to 1/2NS with dextrose and potassium once glucose falls and K+ status is known	Fluids without added K+ replacement once insulin started (risk of hypokalemia)
Burns (Parkland formula)	Lactated Ringer's	D5W; hypertonic saline as sole resuscitation (controversial/limited use)
Hyponatremia (severe/symptomatic)	Hypertonic saline (3%) with slow, monitored correction	Rapid correction of chronic hyponatremia; hypotonic fluids
Hypernatremia	D5W or hypotonic saline (0.45%), correct slowly	Hypertonic or normal saline as maintenance
Cerebral edema / raised ICP	Hypertonic saline or mannitol per protocol	Hypotonic fluids, D5W
Heart failure / renal failure / edematous states	Cautious, restricted volumes; consider albumin (hyperoncotic) in select cases	Large-volume crystalloid boluses
Liver disease with ascites / SBP	Albumin (per protocol, e.g., with SBP or large-volume paracentesis)	Large NS volumes (risk of worsening ascites)

7. Maintenance Fluid Therapy

Holliday-Segar Method (4-2-1 Rule) — Hourly Maintenance Rate

Body Weight	Hourly Rate
First 10 kg	4 mL/kg/hr
Next 10 kg (11-20 kg)	Add 2 mL/kg/hr
Each kg > 20 kg	Add 1 mL/kg/hr

Example: A 70 kg adult = $(10 \times 4) + (10 \times 2) + (50 \times 1) = 40 + 20 + 50 = 110$ mL/hr.

Typical adult maintenance fluid is an isotonic or near-isotonic solution (e.g., 0.9% NaCl or a balanced crystalloid) with potassium added per electrolyte status; historically 0.45% NaCl + dextrose was common but isotonic fluids are now generally preferred, especially in children, due to hyponatremia risk with hypotonic maintenance fluids.

8. Resuscitation Protocols

Sepsis / Septic Shock (Surviving Sepsis Campaign framework)

Initial resuscitation: 30 mL/kg of balanced crystalloid within the first 3 hours for sepsis-induced hypoperfusion, followed by dynamic reassessment of fluid responsiveness (e.g., passive leg raise, stroke volume variation) rather than fixed-volume protocols; vasopressors (norepinephrine first-line) added if hypotension persists after initial fluid resuscitation.

Hemorrhagic Shock / Trauma

Balanced crystalloid used as a temporizing bridge while arranging blood products; permissive hypotension and early balanced blood-product resuscitation (1:1:1 ratio of packed red cells:plasma:platelets in massive transfusion) are favored over large-volume crystalloid resuscitation, which is associated with dilutional coagulopathy and worse outcomes in major hemorrhage.

Diabetic Ketoacidosis (DKA)

Step	Fluid Approach
1. Initial bolus	0.9% NaCl, typically 15-20 mL/kg (or ~1-1.5 L in adults) over the first hour if hypotensive/hypoperfused
2. Ongoing replacement	0.45% or 0.9% NaCl depending on corrected sodium, at a rate to replace the deficit over ~24-48 hrs
3. Potassium	Add K ⁺ once serum K ⁺ <5.2 mEq/L and urine output confirmed; hold insulin if K ⁺ <3.3 mEq/L until replaced
4. Dextrose	Add dextrose (D5 or D10) once glucose falls to ~200-250 mg/dL to prevent hypoglycemia while continuing insulin

Burns (Parkland Formula)

Total fluid (mL) over first 24 hrs = $4 \text{ mL} \times \% \text{TBSA burned} \times \text{body weight (kg)}$, using Lactated Ringer's. Give half of the total in the first 8 hours from time of burn, and the remaining half over the next 16 hours. Titrate to urine output (~0.5 mL/kg/hr in adults, 1 mL/kg/hr in children). Colloid (e.g., albumin) is sometimes added after the first 24 hours in some protocols.

9. Contraindications & Cautions Summary

Fluid	Key Contraindications / Cautions
0.9% NaCl	Large volumes in renal failure, heart failure; risk of hyperchloremic acidosis
Lactated Ringer's	Hyperkalemia, severe hepatic failure, calcium incompatibility with blood/certain drugs (e.g., ceftriaxone)
Plasma-Lyte	Severe hepatic impairment (acetate clearance); cost
D5W	Hypovolemic/hemorrhagic shock, cerebral edema, hyponatremia, uncontrolled hyperglycemia
0.45% NaCl	Acute resuscitation, burns, trauma, hyponatremia, raised ICP
Hypertonic saline	Chronic hyponatremia (rapid correction risk), heart failure, requires central access for concentrated forms
Albumin	Severe fluid overload/heart failure, known hypersensitivity
Starches (HES)	Sepsis, critical illness, renal impairment, coagulopathy - largely avoided/restricted

10. Monitoring & Complications of IV Fluid Therapy

- **Fluid overload:** Peripheral/pulmonary edema, weight gain, worsening oxygenation — monitor daily weights, input/output, lung exam, chest imaging as needed
- **Electrolyte derangement:** Hyper/hyponatremia, hyper/hypokalemia, acid-base disturbance (e.g., hyperchloremic acidosis with large-volume NS) — monitor serial electrolytes
- **Hyperglycemia:** With dextrose-containing fluids, especially in diabetics/critical illness
- **Infusion site complications:** Phlebitis, extravasation (especially with hypertonic or vesicant solutions), infection
- **Osmotic demyelination syndrome:** With overly rapid correction of chronic hyponatremia using hypertonic saline
- **Dilutional coagulopathy:** With large-volume crystalloid resuscitation in hemorrhage
- **Renal effects:** Some evidence links large-volume, high-chloride fluid (NS) with increased AKI risk versus balanced crystalloids in critically ill patients

11. Quick Reference Comparison Table

Fluid	Tonicity	Best For	Main Risk
0.9% NaCl	Isotonic	General resuscitation, hyponatremia, blood product compatibility	Hyperchloremic acidosis in large volumes
Lactated Ringer's	~Isotonic	Surgery, trauma, burns	Hyperkalemia risk, blood-line incompatibility
Plasma-Lyte	Isotonic	ICU/sepsis, renal protection	Cost, hepatic acetate clearance
D5W	Isotonic bag / physiologically hypotonic	Free water deficit, hypernatremia	Poor volume expander, hyponatremia, hyperglycemia
0.45% NaCl	Hypotonic	Maintenance, hypernatremia	Cerebral edema, hyponatremia
3% Hypertonic saline	Hypertonic	Severe hyponatremia, raised ICP	Osmotic demyelination if corrected too fast
Albumin	Iso/hyperoncotic	SBP, hepatorenal syndrome, refractory edema	Cost, no clear mortality benefit generally

Fluid	Tonicity	Best For	Main Risk
Starches (HES)	Colloid	Largely avoided now	AKI, coagulopathy

12. IV Drip Rate & Flow Rate Calculation

IV fluids can be delivered by gravity (drip counting) or by infusion pump (mL/hr). Manual drip counting requires knowing the **drop factor** of the giving set, which is printed on the IV administration set packaging.

Drop Factor (Giving Set Types)

Set Type	Drop Factor (gtt/mL)	Typical Use
Macro drip (standard)	10 gtt/mL	Some adult sets, larger-volume fluids
Macro drip (standard)	15 gtt/mL	Most common adult IV set
Macro drip (standard)	20 gtt/mL	Some adult sets (check packaging)
Micro drip (pediatric/precise)	60 gtt/mL	Pediatrics, neonates, critical/precise low-volume infusions
Blood administration set	10-15 gtt/mL	Packed red cells, blood products

Always confirm the drop factor printed on the specific administration set in use — it varies by manufacturer.

Core Formulas

What You Need	Formula
Flow rate (mL/hr)	Total volume (mL) ÷ Time (hr)
Drops per minute (gtt/min)	(Volume in mL × Drop factor [gtt/mL]) ÷ Time in minutes
Drops per minute (from mL/hr)	(mL/hr × Drop factor) ÷ 60
Time to complete infusion (hr)	Total volume (mL) ÷ Flow rate (mL/hr)
mL/hr from a pump order given in gtt/min	(gtt/min × 60) ÷ Drop factor

Worked Examples

Example 1 — Flow rate in mL/hr: Infuse 1000 mL over 8 hours.

Flow rate = $1000 \div 8 = 125$ mL/hr.

Example 2 — Drops per minute with a macro drip (15 gtt/mL) set: Infuse 1000 mL over 8 hours (480 minutes).

gtt/min = $(1000 \times 15) \div 480 = 15000 \div 480 = \sim 31$ gtt/min.

Example 3 — Drops per minute from a known mL/hr rate: Pump rate ordered as 125 mL/hr, using a 20 gtt/mL set (for manual/gravity backup).

gtt/min = $(125 \times 20) \div 60 = 2500 \div 60 = \sim 42$ gtt/min.

Example 4 — Micro drip for a pediatric/precise order: Infuse 250 mL over 5 hours using a 60 gtt/mL micro drip set.

gtt/min = $(250 \times 60) \div 300 = 15000 \div 300 = 50$ gtt/min (note: with a 60 gtt/mL set, gtt/min always numerically equals mL/hr — a useful shortcut).

Example 5 — Rapid bolus: Give a 500 mL crystalloid bolus over 30 minutes via macro drip (15 gtt/mL).

gtt/min = $(500 \times 15) \div 30 = 7500 \div 30 = 250$ gtt/min (in practice, boluses this fast are usually run via pump or pressure bag, not manually counted, since >60 gtt/min is hard to count accurately by eye).

Quick-Reference: mL/hr to gtt/min by Set Type

mL/hr	10 gtt/mL	15 gtt/mL	20 gtt/mL	60 gtt/mL (micro)
50	8	13	17	50

mL/hr	10 gtt/mL	15 gtt/mL	20 gtt/mL	60 gtt/mL (micro)
75	13	19	25	75
100	17	25	33	100
125	21	31	42	125
150	25	38	50	150
200	33	50	67	200

Practical Tips

- Count drops for a full 60 seconds (or count for 15 sec x 4) for accuracy, watching the drip chamber.
- With a 60 gtt/mL microdrip set, gtt/min = mL/hr numerically — convenient for pediatric or tightly titrated infusions.
- Whenever available, use an infusion pump (programmed in mL/hr) for accuracy, especially for vasoactive drugs, hypertonic saline, insulin, and pediatric/neonatal fluids — manual gravity drip is imprecise and affected by patient position, tubing height, and venous pressure.
- Recalculate and re-time the infusion periodically, since gravity flow rates can drift.
- Always double-check the ordered rate, total volume, and infusion time against the prescription before setting up the drip.

This summary is intended as a study/reference aid and does not replace institutional protocols, drug references, or clinical judgment for individual patients.